Programme Theory

Programme Theory (v12)

Sign-off: PENDING (HITL-4)

Across the accumulated evidence, undergraduate clinical-reasoning interventions appear to work less through a simple analytic-versus-non-analytic superiority effect than through the fit between learner knowledge, case/task demand, and instructional support. Case-based exposure, examples, feedback, self-explanation, contrast, and explicit reasoning instruction become generative when they help students build usable schemas, recognise diagnostically relevant features, compare alternatives, revise errors, and calibrate when to trust or check first impressions. The same resources can fail or backfire when learners lack prerequisite illness-script knowledge, when prompts are ambiguous or too demanding, when feedback is too thin, when modelling is tacit or cognitively remote, or when self-directed resources are optional and weakly taken up.

Context 1: Novices with little prior domain exposure facing confusable diagnostic categories

> *When (C) undergraduate novices have little or no prior experience in a diagnostic domain and the categories are visually or clinically confusable, (Mresource) Combined Diagnostic Instruction, Schema-guided diagnostic reasoning, and Diagnostic reasoning scaffolds offer explicit feature cues, contrasts, annotated cases, or schema-based distinguishing relations; (Mresponse) learners attend to diagnostically decisive features, compare similar alternatives, organise knowledge into usable diagnostic structures, and temper premature familiarity-based closure; leading to (O) improved diagnostic accuracy, structured knowledge, and better performance on taught and some untaught cases.*

This configuration refines the theory by showing that novices can benefit from analytic and non-analytic resources together when the instruction is bounded and makes diagnostic relations visible. ECG, cardiac auscultation, cytology, and general diagnostic tasks suggest that contrast, schema grouping, feature annotation, and explicit combined strategy instruction can help learners develop early illness scripts and more discriminating pattern recognition rather than merely applying a checklist.

Supporting studies: S003, S005, S017, S021

Conflicting evidence: The contradiction is context-sensitive. Feature analysis and combined strategy instruction were positive when they clarified confusable categories, but negative when the feature table or combined prompt was unclear, too exhaustive, or allowed novices to misidentify features. Thus contrast and feature analysis are not inherently beneficial; they require bounded, interpretable scaffolding matched to novice knowledge.

Context 2: Novices or underprepared learners given unbounded analytic, metacognitive, or bias-correction strategies

> *When (C) learners have underdeveloped illness scripts, little clinical experience, or insufficient domain knowledge, (Mresource) unguided free-choice prompts, speeded single-diagnosis instructions, cognitive forcing seminars, tacit clinician reasoning, or ambiguous combined analytic/non-analytic prompts offer reasoning strategies without enough schema support; (Mresponse) learners engage only partially in reflection, become confused, rely on superficial feature-counting or common diagnoses, or fail to perceive the reasoning pathway; leading to (O) no improvement, lower diagnostic accuracy, limited error prevention, or barriers to learning clinical reasoning.*

This section explains why strategy coaching can fail. The resource may ask for analytic checking, second-diagnosis search, or bias monitoring, but the learner response depends on having enough organised knowledge to use the strategy. Where that knowledge is absent, students may not know what to compare, which features matter, or how to translate an expert's automated reasoning into their own developing schema.

Supporting studies: S005, S016, S021, S024, S028

Conflicting evidence: This does not contradict evidence that reasoning instruction can help. It specifies the negative context: strategy instruction backfires or becomes inert when it is abstract, tacit, ambiguous, or cognitively demanding before learners have usable diagnostic schemas. The same broad driver is positive in contexts where instruction is concrete, case-linked, and scaffolded.

Context 3: Learners with theoretical knowledge but limited patient exposure who can use structured reflection and comparison

> *When (C) students possess some theoretical or domain knowledge but have limited patient exposure or still-developing illness scripts, (Mresource) Clinical reasoning scaffolding and Diagnostic reasoning scaffolds provide structured reflection, absent-feature prompts, analytic tables, similar-case series, decision-point questions, and feedback; (Mresponse) learners compare findings across cases, consider alternatives and conflicting evidence, reorganise illness scripts, strengthen case representations, and practise retrieval; leading to (O) improved diagnostic transfer, reasoning competence, longer-term retention, or higher diagnostic accuracy.*

This configuration is central to the refined programme theory. Students who know some content but lack rich clinical scripts are able to use structured comparison and feedback to reorganise knowledge around diagnostic decisions. The resource does not simply add information; it changes how learners relate findings, alternatives, and prior knowledge.

Supporting studies: S016, S018, S019, S023, S027

Conflicting evidence: The positive pathway is moderated by task demand. Structured comparison can reduce performance immediately when it is stepwise and cognitively heavy, similar-case learning can be only partly exploited when misconceptions remain unresolved, and analytic tables

add little when a case is typical enough for pattern recognition to be sufficient. The contradiction is therefore between productive difficulty and overload or redundancy.

Context 4: Students working with less familiar but sufficiently bounded clinical cases

> *When (C) learners encounter less familiar cases that are challenging but still within reach of their prior knowledge, (Mresource) self-explanation diagnostic case solving, pathway review, and Feedback- and Example-Driven Reasoning Gain resources provide prompts, definitions, examples, and opportunities to compare one's reasoning with examples; (Mresponse) learners actively elaborate, generate biomedical inferences, link clinical and biomedical knowledge, revise knowledge representations, and build more coherent disease models; leading to (O) improved delayed diagnostic performance, near transfer, and in some conditions far-transfer gains.*

Self-explanation appears most generative when the case is unfamiliar enough to require sense-making but not so unsupported that learners construct errors unchecked. The mechanism response is not merely talking aloud; it is active reconstruction of explanatory links that later support retrieval and diagnosis.

Supporting studies: S013, S020, S025, S026

Conflicting evidence: The same self-explanation resource can produce paraphrases, errors, or little advantage when learners' specific knowledge is too thin, when examples contain uncorrected peer errors, or when expert reasoning skips steps that students cannot relate to their own knowledge. Near-peer or cognitively congruent examples help resolve this by being more developed than students' reasoning but still interpretable.

Context 5: Learners with poor, incomplete, or error-prone knowledge facing complex diagnostic environments

> *When (C) learners' prior knowledge is fragmented or error-prone and the diagnostic environment is complex, (Mresource) case-based worked examples, structured diagnostic schemas, and elaborated feedback provide worked diagnostic processes and explanations of errors; (Mresponse) learners reduce extraneous cognitive load, identify errors, understand correct-solution logic, and allocate cognitive resources to schema construction; leading to (O) more effective diagnostic knowledge acquisition and conceptual learning.*

For underprepared learners, support intensity matters. Worked examples and elaborated feedback can make complex reasoning tractable by showing the diagnostic process and explaining why errors matter. This supports schema construction before learners are expected to perform fully independent reasoning.

Supporting studies: S006, S018, S022

Conflicting evidence: Thin feedback or excessive task difficulty changes the mechanism response. Right-or-wrong feedback, official diagnosis feedback alone, or fixed timed challenges can trigger

superficial deduction, unresolved misconceptions, frustration, or failure to debug reasoning. Thus feedback is beneficial when explanatory and schema-linked, but insufficient when it merely confirms correctness.

Context 6: Advanced or near-practice students facing authentic decision making, uncertainty, and communication demands

> *When (C) students are advanced, near licensure, or approaching transition to clinical responsibility, (Mresource) Clinical reasoning scaffolding through analytic-reasoning tables, simulated ward scenarios, structured communication prompts, and realistic decision cases offers authentic but bounded decision practice; (Mresponse) learners consider alternatives, manage conflicting findings, use emerging clinical experience, feel more comfortable communicating with seniors, and adapt structured tools to practice; leading to (O) higher diagnostic accuracy in suitable cases, increased subjective preparedness, and improved decision-making confidence.*

At this stage, students can often use analytic scaffolds and authentic simulations because they have enough clinical knowledge to connect prompts with action. The mechanism response includes both cognitive reasoning and confidence calibration for practice transitions.

Supporting studies: S015, S019, S022

Conflicting evidence: Authenticity is not automatically productive. When students lack prior prioritisation or multitasking experience, pager interruptions, acutely unwell patients, or passive shadowing can trigger stress, abandonment of structured assessment, or feeling cocooned from responsibility. Authenticity helps when it is scaffolded into graduated responsibility rather than mere exposure.

Context 7: Learners developing non-analytic pattern recognition through repeated case exposure

> *When (C) students have limited access to common presentations or need repeated exposure to diagnostic variation, (Mresource) Structured self-directed practice, simulation and case-based practice, virtual patients, EHR case pools, and similar-case series provide repeated real or simulated cases, labelled exemplars, longitudinal scenarios, and feedback; (Mresponse) learners accumulate exemplar familiarity, form more secure case representations, observe test-of-time or test-of-treatment reasoning, and develop illness scripts and pattern recognition; leading to (O) improved reasoning competence, confidence, and diagnostic performance, especially for previously seen or related cases.*

This configuration captures the non-analytic side of the programme theory. Repeated exposure does not simply increase time-on-task; it can supply the experience base from which pattern recognition and illness scripts develop. Its strongest effects appear when exposure is meaningful, varied, and linked to feedback or diagnostic decision points.

Supporting studies: S005, S011, S014, S018, S022, S023

Conflicting evidence: Exposure alone is fragile. Optional access can produce limited uptake, similar-case exposure may not correct misconceptions without sufficient feedback, and difficult timed cases may frustrate novices. Non-analytic development therefore depends on designed exposure rather than mere availability of cases.

Context 8: Students engaging with self-directed or low-stakes digital learning resources

> *When (C) learners are offered optional, self-paced, or formative resources with low external incentive, (Mresource) Structured self-directed practice and e-seminar resources provide repeatable cases, questions, feedback, and flexible access; (Mresponse) some learners use feedback and retrieval opportunities to correct errors, but others engage minimally or do not revise because the activity carries little consequence; leading to (O) variable uptake, weak overall retention in some settings, but item-level learning and improved long-term reasoning where repeated testing and feedback are actually used.*

This section adds an uptake mechanism to the theory. A resource can be well designed cognitively but still fail to generate outcomes if learners do not use it sufficiently. Conversely, when self-paced practice includes repeated case-based questions and immediate explanatory feedback, learners can benefit through retrieval and feedback processing.

Supporting studies: S011, S014, S027

Conflicting evidence: The apparent contradiction between popularity or usefulness and restricted learning effect is resolved by distinguishing availability from engagement. Self-directed resources can support reasoning practice, but optional or non-credit formats may not reliably trigger sustained effort.

Context 9: Learners observing modelled reasoning from clinicians, experts, peers, or near-peers

> *When (C) students need to see how diagnostic reasoning is performed but differ from the model in expertise, (Mresource) Clinical reasoning scaffolding, explicit physician-reasoning models, and self-explanation examples provide visible reasoning pathways, prompts, and modelled diagnostic arguments; (Mresponse) learners understand thought processes, compare their reasoning with the model, elaborate or complete their own arguments, and articulate reasoning during work-up; leading to (O) improved reasoning ability, diagnostic performance, and reasoning efficiency when the model is concrete and cognitively congruent.*

Modelling works when it makes reasoning visible at a level the learner can process. Real physician reasoning, prompted consultation frameworks, and resident near-peer self-explanations can help students translate observed reasoning into their own developing schemas.

Supporting studies: S023, S025, S026, S028

Conflicting evidence: Expertise distance explains conflicting findings. Tacit clinician reasoning, expert explanations with skipped abstract steps, or peer examples containing irrelevant details and uncorrected errors may trigger little learning or difficulty grasping the reasoning. Near-peer models and explicit prompts reduce this gap by making the reasoning pathway interpretable.

Boundary statement

This automated realist synthesis composes a provisional programme theory from the supplied demi-regularities, conceptual entities, contradictions, and CMOC statements. It does not claim that these are final causal truths, does not adjudicate the normative superiority of analytic versus non-analytic reasoning, and does not replace reviewer interpretation. Final interpretive judgement, weighting of evidence, and normative theory choice remain with the human researchers.